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HACK THE BOX SHERLOCK

Dream Job-2

1. According to MITRE ATT&CK, what previously known malware does DRATzarus share similarities with?

   DRATzarus

   DRATzarus is a remote access tool (RAT) that has been used by Lazarus Group to target the defense and aerospace organizations globally since at least summer 2020. DRATzarus shares similarities with Bankshot, which was used by Lazarus Group in 2017 to target the Turkish financial sector.[1]

   https://attack.mitre.org/software/S0694/
   In the website of MITRE ATT&CK specifically for DRATzarus, it states the answer that DRATzarus malware is similar to **Bankshot**.

2. Which Windows API function does DRATzarus use to detect the presence of a debugger?

Enterprise	T1622	Debugger Evasion	DRATzarus can use `IsDebuggerPresent` to detect whether a debugger is present on a victim.[1]

   Scrolling down the website from question number 1, we can see the answer is **IsDebuggerPresent**, which is used to check whether a debugger is present on the victim's system.

3. Torisma is another piece of malware used by the Lazarus Group. According to MITRE, it has encrypted its C2 communications using XOR and which other method?

Enterprise	T1573	.001	Encrypted Channel: Symmetric Cryptography	Torisma has encrypted its C2 communications using XOR and VEST-32.[1]

   https://attack.mitre.org/software/S0678/
   In the screenshot above, we can see that the answer is there on the MITRE ATT&CK website for Torisma specifically, and I also have given the link and the answer is using **VEST-32**.

4. Which packing method has been used to obfuscate Torisma?

Enterprise	T1027	.002	Obfuscated Files or Information: Software Packing	Torisma has been packed with lz4 compression.[1]
		.013	Obfuscated Files or Information: Encrypted/Encoded File	Torisma has been Base64 encoded and AES encrypted.[1]

Still within the same website as number 3, scrolling down we can see the answer to obfuscating the files in software packing and it is used by a compression algorithm called **lz4 compression**.

5. Analyze the provided ISO file and identify the executable contained within it?

```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/DreamJob2]
└─$ 7z x BAE_HPC_SE.iso

7-Zip 25.01 (x64) : Copyright (c) 1999-2025 Igor Pavlov : 2025-08-03
 64-bit locale=en_SG.UTF-8 Threads:4 OPEN_MAX:1024, ASM

Scanning the drive for archives:
1 file, 10786816 bytes (11 MiB)

Extracting archive: BAE_HPC_SE.iso
--
Path = BAE_HPC_SE.iso
Type = Iso
Physical Size = 10786816
Comment =
{
VolumeSpaceSize: 10786816
VolumeSetSize: 1
VolumeSequenceNumber: 1
}
Created = 2020-06-05 15:04:49.00
Modified = 2020-06-05 15:04:49.00

Everything is Ok

Files: 2
Size:       10734110
Compressed: 10786816
```

We can see from the screenshot above, there are 2 files inside this ISO, and then after doing "ls -la" on that same directory we can see what the files are and we can see an executable like shown in the screenshot below.

```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/DreamJob2]
└─$ ls -la
total 47800
drwxrwxr-x 2 rafael rafael     4096 Mar 23 21:46 .
drwxrwxr-x 7 rafael rafael     4096 Mar 22 23:22 ..
-rw-rw-r-- 1 rafael rafael  1283075 Mar  5  2025 17.dotm
-rw-rw-r-- 1 rafael rafael 10786816 Mar  5  2025 BAE_HPC_SE.iso
-rw-rw-r-- 1 rafael rafael   226846 Jun  5  2020 BAE_HPC_SE.pdf
-rw-rw-r-- 1 rafael rafael     1051 Nov 20 16:19 DANGER.txt
-rw-rw-r-- 1 rafael rafael 12410660 Nov 20 16:19 DANGER.zip
-rw-rw-r-- 1 rafael rafael 12411514 Mar 22 23:22 DreamJob-2.zip
-rw-rw-r-- 1 rafael rafael 10507264 Jun  5  2020 InternalViewer.exe
-rw-rw-r-- 1 rafael rafael  1293824 Mar  5  2025 Salary_Lockheed_Martin_job_opportunities_confidential.doc
```

We can see that the answer I blocked, it is **InternalViewer.exe**

6. The executable found in the previous question was renamed. Can you identify its original name?

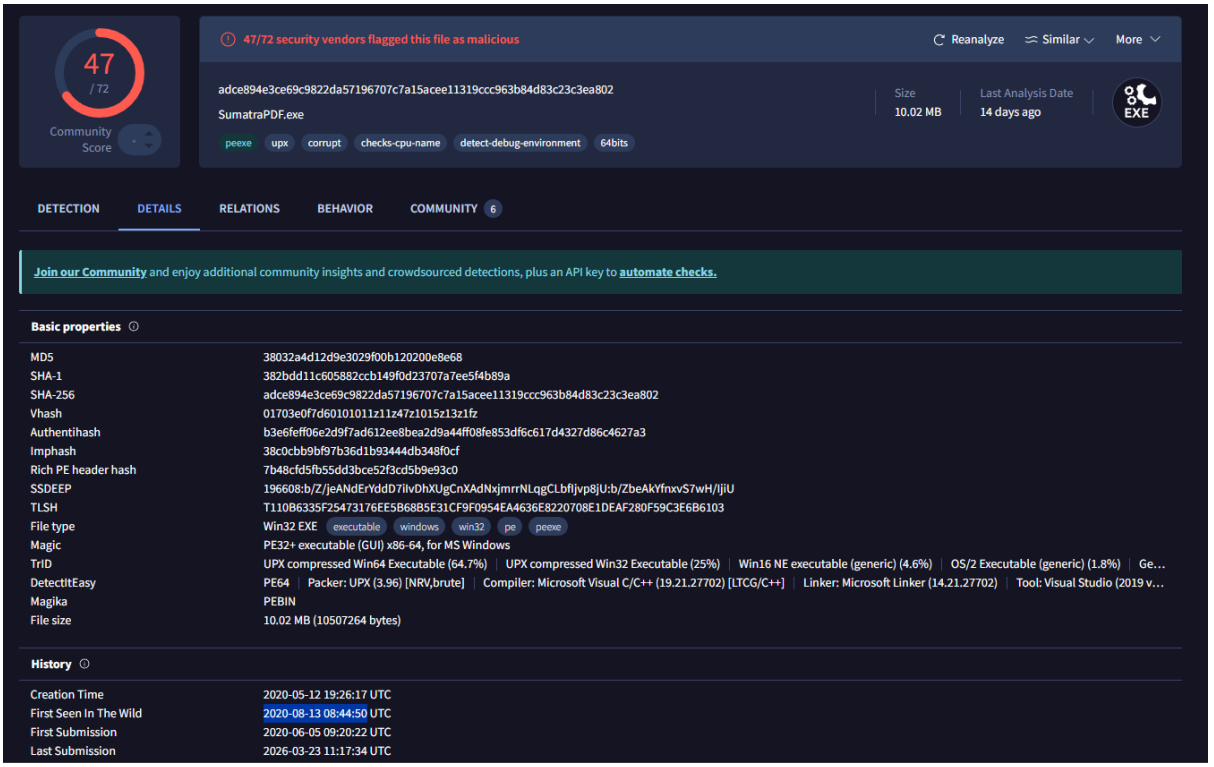
```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/DreamJob2]
└─$ exiftool InternalViewer.exe
ExifTool Version Number         : 13.25
File Name                       : InternalViewer.exe
Directory                       : .
File Size                       : 11 MB
File Modification Date/Time     : 2020:06:05 15:00:44+08:
File Access Date/Time           : 2026:03:23 21:52:17+08:
File Inode Change Date/Time     : 2026:03:23 21:46:39+08:
File Permissions                : -rw-rw-r--
File Type                       : Win64 EXE
File Type Extension             : exe
MIME Type                       : application/octet-strea
Machine Type                    : AMD AMD64
Time Stamp                      : 2020:05:13 03:26:17+08:
Image File Characteristics      : Executable, Large addre
PE Type                         : PE32+
Linker Version                  : 14.21
Code Size                       : 10465280
Initialized Data Size           : 45056
Uninitialized Data Size         : 34689024
Entry Point                     : 0x2b10580
OS Version                      : 6.0
Image Version                   : 0.0
Subsystem Version               : 6.0
Subsystem                       : Windows GUI
File Version Number             : 3.2.0.0
Product Version Number          : 3.2.0.0
File Flags Mask                 : 0x0000
File Flags                      : (none)
File OS                         : Windows NT 32-bit
Object File Type                : Executable application
File Subtype                    : 0
Language Code                   : English (U.S.)
Character Set                   : Windows, Latin1
File Description                : SumatraPDF
File Version                    : 3.2
Legal Copyright                 : Copyright 2006-2020 all
Original File Name              : SumatraPDF.exe
Product Name                    : SumatraPDF
Product Version                 : 3.2
Company Name                    : Krzysztof Kowalczyk
```

As can be seen blocked by the screenshot above, the Original File Name can be spotted and turns out to be **SumatraPDF.exe**. I used the tool exiftool to see metadata about a particular file. The attacker renamed it to make it blend in and not suspicious, while adding in malicious code. This tactic is also included in the APT (Advanced Persistent Threat) attack by the Lazarus group.

7. According to VirusTotal, when was the EXE from the previous question First Seen In The Wild?(UTC)

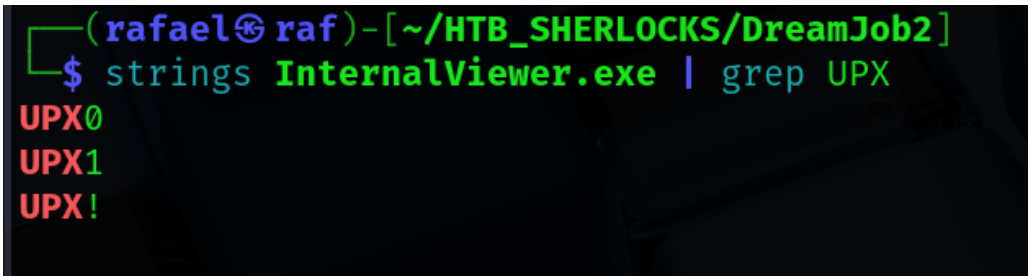
```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/DreamJob2]
└─$ sha1sum InternalViewer.exe
382bdd11c605882ccb149f0d23707a7ee5f4b89a  InternalViewer.exe
```

To make it simpler, I just hashed the executable file using the sha1sum tool, and there I got the hashed string for InternalViewer.exe. And then I copy pasted it to VirusTotal.



Then, like asked in the question I found the UTC time, when it was first seen in the wild, so the answer is **2020-08-13 08:44:50**

8. What packer was used to pack the executable from Question 6? (Full name)



I used the tool strings for the InternalViewer.exe, but I tried UPX, which is a standard packer commonly used for practical malware analysis. It stands for **Ultimate Packer for eXecutables**, and used for packer and compressor for all files across operating systems.

9. What is the full URL found within the macro in the document Salary_Lockheed_Martin_job_opportunities_confidential.doc?

```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/DreamJob2]
└─$ strings Salary_Lockheed_Martin_job_opportunities_confidential.doc | grep "https"
https://markettrendingcenter.com/lk_job_oppor.docx
```

The doc file is very long, therefore I searched again using the same format as the previous question but this time since it is asking for a url, I filtered it with https, and the answer is found to be **https://markettrendingcenter.com/lk_job_oppor.docx**

10. Who is the author of the document Salary_Lockheed_Martin_job_opportunities_confidential.doc?

```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/DreamJob2]
└─$ file Salary_Lockheed_Martin_job_opportunities_confidential.doc
Salary_Lockheed_Martin_job_opportunities_confidential.doc: Composite Document File V2 Document, Little Endian, Os: Windows, Version 10.0, Code page: 1252, Author: Mickey, Template: Normal.dotm, Last Saved By: Challenger, Revision Number: 83, Name of Creating Application: Microsoft Office Word, Total Editing Time: 37:00, Create Time/Date: Fri Apr 24 03:18:00 2020, Last Saved Time/Date: Mon Oct 18 13:06:00 2021, Number of Pages: 1, Number of Words: 4, Number of Characters: 29, Security: 0
```

I just used the file command for the doc file and then we can see all the necessary information about the file, including the author of the document, who is **Mickey.**

11. Who last modified the above document?

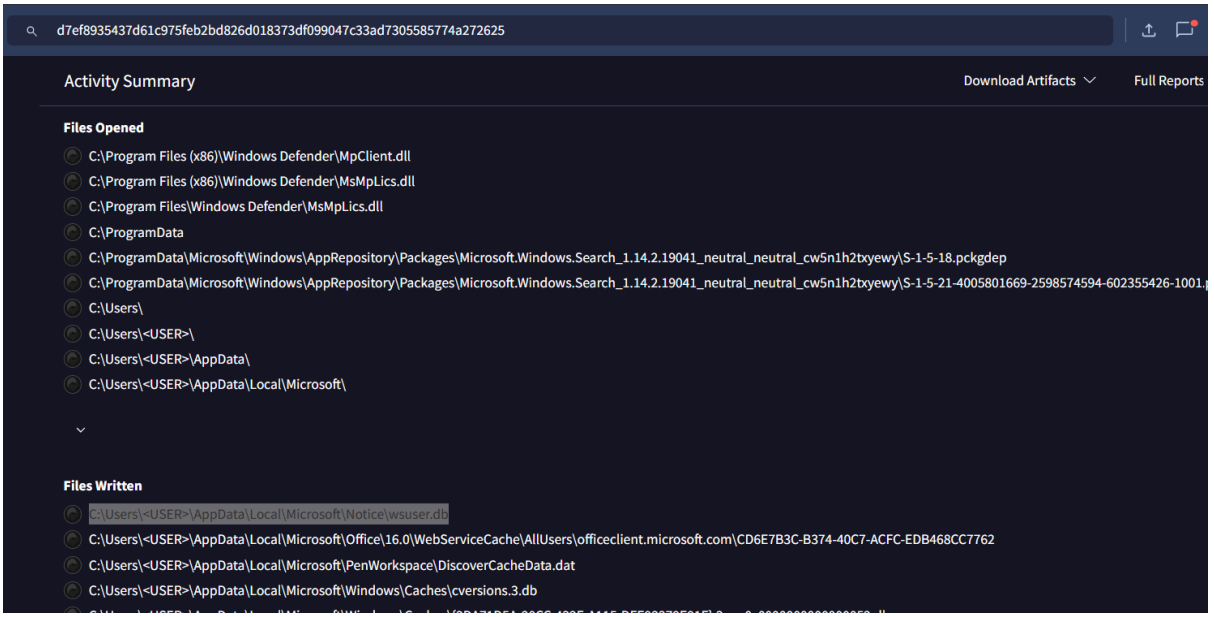
```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/DreamJob2]
└─$ file Salary_Lockheed_Martin_job_opportunities_confidential.doc
Salary_Lockheed_Martin_job_opportunities_confidential.doc: Composite Document File V2 Document, Little Endian, Os: Windows, Version 10.0, Code page: 1252, Author: Mickey, Template: Normal.dotm, Last Saved By: Challenger, Revision Number: 83, Name of Creating Application: Microsoft Office Word, Total Editing Time: 37:00, Create Time/Date: Fri Apr 24 03:18:00 2020, Last Saved Time/Date: Mon Oct 18 13:06:00 2021, Number of Pages: 1, Number of Words: 4, Number of Characters: 29, Security: 0
```

Still from the same output of the previous question number 10, we can see the Last Saved By parameter and I highlighted the answer, which is to be **Challenger**

12. Analyze the "17.dotm" document. What is the directory where a suspicious folder was created? (Format: Give the path starting immediately after <USER>. Please pay attention to placeholder.)

```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/DreamJob2]
└─$ sha1sum 17.dotm
6a82ebf8a6e868323b1de30a81cd6658ce4c31b6  17.dotm
```

I took the hash value for the file of 17.dotm. Then I put it on Virus Total again to see the behavior section and see what are the files created and written.



As highlighted on the screenshot above, there was a suspicious folder leading to a database file called wsuser.db, so I conclude that the answer is **\AppData\Local\Microsoft\Notice**

13. Which suspicious file was checked for existence in that directory?

Still using the same screenshot as the last question and already answered from the previous question, which text is colored blue the answer is **wsuser.db**